

Training of a DDPM:

* Given a i/p data point x_0 , from the true data dist

* Repeat untill Convergence :

pick a $t \sim \text{Uniform} \{0, \dots, T\}$

$$\tilde{x}_t^m \sim N(x_t; \sqrt{\bar{\alpha}_t} x_0, (1 - \bar{\alpha}_t) I)$$

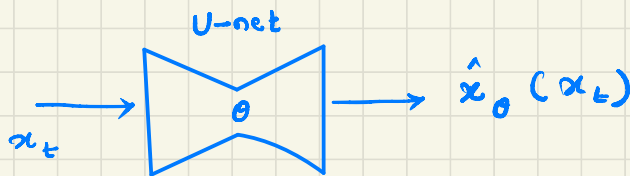
$$\tilde{x}_t^m = \bar{\alpha}_t x_0 + (1 - \bar{\alpha}_t) \cdot \epsilon_t, \quad \epsilon_t \sim N(0, I)$$

$$\theta^{k+1} \leftarrow \theta^k - \beta \nabla_{\theta} J_0(q)$$

$$\text{where } J_0(q) = \sum_{i=1}^m \|\hat{x}_{\theta}(\tilde{x}_t^m) - x_0\|_2^2$$

$m \in \{0, \dots, T\}$ } the outer
diff time steps } IE

Forward pass
Compute $\hat{x}_{\theta}(x_t)$



back prop

Simple &
elegant
training.

Note: Do not forget the 1st Term that needs to be added

Inference / Sampling in DDPM:

* To get a sample $x_0 \sim p_{x_0}$, we need to traverse through the backward @ decoding process

$$x_T \rightarrow x_{T-1} \rightarrow \dots \rightarrow x_0$$

=> we need to sample iteratively from decoding dist

$$x_{t-1} \sim p_\theta(x_{t-1} | x_t) \quad \forall t = T \dots 1$$

$$x_T \sim N(0, I)$$

$x_1, \dots, x_T$ are fixed $[0, 1]$
 $T \approx 1000$

$$x_{T-1} \sim p_\theta(x_{T-1} | x_T)$$

$$x_{T-2} \sim p_\theta(x_{T-2} | x_{T-1})$$

$\vdots$

$$x_0 \sim p_\theta(x_0 | x_1)$$

$$p_\theta(x_{t-1} | x_t) = N(x_{t-1}; \mu_\theta^*(x_t), \sigma_q^2 I)$$

$$x_{t-1} = \mu_\theta^*(x_t) + \sigma_q^2 \cdot \epsilon \quad \epsilon \sim N(0, I)$$

when we are regressing over x_0

$$\mu_\theta^*(x_t) = \frac{(1 - \bar{\alpha}_{t-1}) \sqrt{\alpha_t}}{1 - \bar{\alpha}_t} x_t + \frac{(1 - \alpha_t) \sqrt{\bar{\alpha}_{t-1}}}{1 - \bar{\alpha}_t} \hat{x}_\theta(x_t)$$

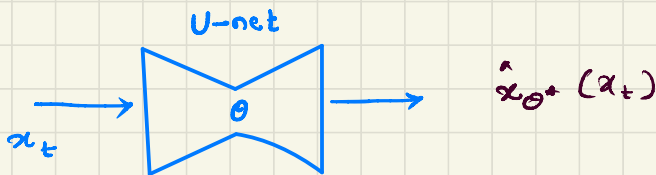
Alg 0

start with $x_T \sim N(0, I)$

for $t = T, T-1, \dots, 1$

$$x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, \quad \epsilon \sim N(0, I)$$

get $\hat{x}_{\theta^*}(x_t)$ from the NN.



$$x_{t-1} = \mu_{\theta^*}(x_t) + \sigma_q^2 \cdot \epsilon \quad \epsilon \sim N(0, I)$$

where

$$\mu_{\theta^*}(x_t) = \frac{(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} \sqrt{\alpha_t} x_t + \frac{(1 - \alpha_t) \sqrt{\bar{\alpha}_{t-1}}}{1 - \bar{\alpha}_t} \hat{x}_{\theta^*}(x_t)$$

return x_0 : A new generated sample.

To sample in a DDPM, we need to perform T number of forward processes. Through the Trained NN (θ^*)

Alternate Interpretations of DDPM:

a) DDPM as regression over added "noise"

Recall the forward process of DDPM,

$$x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon_t, \quad \epsilon_t \sim N(0, 1)$$

Rearranging the above terms,

$$x_0 = \frac{x_t - \sqrt{1 - \bar{\alpha}_t} \epsilon_t}{\sqrt{\bar{\alpha}_t}} \rightarrow \textcircled{1}$$

ϵ_t : Noise added to x_0 to convert it to x_t

Recall the consistency term in the ELBO

$$\|h_0 - h_q\|_2^2$$

h_q : mean of $q(x_{t-1} | x_t, x_0)$

h_0 : mean of $p_0(x_{t-1} | x_t)$

Represent h_q in terms of x_t & ϵ_t

$$h_q(x_t, x_0) = \frac{\sqrt{\alpha_t} (1 - \bar{\alpha}_{t-1}) x_t + \sqrt{\bar{\alpha}_{t-1}} (1 - \alpha_t) x_0}{1 - \bar{\alpha}_t}$$

Replacing x_0 above using eqn ①

$$\mu_q(x_t, x_0) = \sqrt{\alpha_t} (1 - \bar{\alpha}_{t-1}) x_t + \bar{\alpha}_{t-1} (1 - \alpha_t) \left(\frac{x_t - \sqrt{1 - \bar{\alpha}_t} \epsilon_t}{\sqrt{\alpha_t}} \right)$$

$$1 - \bar{\alpha}_t$$

⋮

skipping algebra.

$$\mu_q(x_t, \epsilon_t) = \frac{1}{\sqrt{\alpha_t}} \cdot x_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t} \sqrt{\alpha_t}} \cdot \epsilon_t$$

$$\mu_0(x_t) = \frac{1}{\sqrt{\alpha_t}} x_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t} \sqrt{\alpha_t}} \hat{\epsilon}_0(x_t)$$

earlier the consistency term

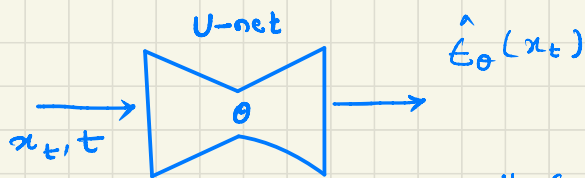
$$= \frac{1}{2 \sigma_q^2} \|\mu_q - \mu_0\|_2^2$$

Substituting the values obtained earlier

$$= \frac{1}{2 \sigma_q^2} \frac{(1 - \alpha_t)^2}{(1 - \bar{\alpha}_t) \alpha_t} \left[\|\epsilon_t - \hat{\epsilon}_0(x_t)\|_2^2 \right]$$

$$\propto \|\epsilon_t - \hat{\epsilon}_0(x_t)\|_2^2$$

Regression over the added noise



$\| \epsilon_t - \hat{\epsilon}_\theta(x_t) \|_2^2$
 Regressor on the added noise

b) DDPM as score-predictors:

Suppose $t \sim N(t; \mu_t, \Sigma_t) = p(t)$

Tweedie's Formula

$$E[\mu_t | t] = t + \Sigma_t \cdot \nabla_t \log p(t)$$

where $\nabla_t \log p(t)$ defined as the "score-func"

* it tells us $\rightarrow$ of log likelihood traverse

Recall, that in a DDPM

$$q(x_t | x_0) = N(x_t; \sqrt{\bar{\alpha}_t} \cdot x_0, (1 - \bar{\alpha}_t) I)$$

apply the Tweedie's formula to above.

$$E[\mu | x_t] = x_t + (1 - \bar{\alpha}_t) \nabla_{x_t} \log p(x_t)$$

The best estimate for $E[\mu | x_t]$ is the true mean

which is $\sqrt{\bar{\alpha}_t} x_0$

$$\therefore \sqrt{\bar{\alpha}_t} x_0 = x_t + (1 - \bar{\alpha}_t) \nabla_{x_t} \log p(x_t)$$

Rearranging the terms.

$$x_0 = \frac{x_t + (1 - \bar{\alpha}_t) \nabla_{x_t} \log p(x_t)}{\sqrt{\bar{\alpha}_t}}$$

Now expressing the consistency terms in terms of series

$$h_2(x_t, x_0) = \frac{\sqrt{\alpha_t} (1 - \bar{\alpha}_{t-1}) x_t + \sqrt{\bar{\alpha}_{t-1}} (1 - \alpha_t) x_0}{1 - \bar{\alpha}_t}$$

Substituting x_0 obtained earlier.

$$= \frac{\sqrt{\alpha_t} (1 - \bar{\alpha}_{t-1}) x_t + \sqrt{\bar{\alpha}_{t-1}} (1 - \alpha_t) \frac{x_t + (1 - \bar{\alpha}_t) \nabla_{x_t} \log p(x_t)}{\sqrt{\bar{\alpha}_t}}}{1 - \bar{\alpha}_t}$$

: skipping algebra

$$h_2 = \frac{1}{\sqrt{\alpha_t}} x_t + \frac{1 - \alpha_t}{\sqrt{\alpha_t}} \cdot \nabla_{x_t} \log p(x_t)$$

$$h_0 = \frac{1}{\sqrt{\alpha_t}} x_t + \frac{1 - \alpha_t}{\sqrt{\alpha_t}} \cdot s_0(x_t)$$

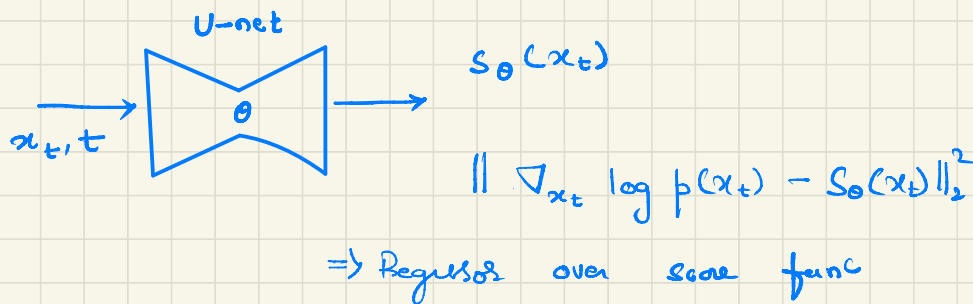
$s_0(x_t)$: predicted score using NN.

now the consistency term

$$\propto \| \mu_t - \mu_0 \|_2^2$$

$$\propto \| \nabla_{x_t} \log p(x_t) - s_0(x_t) \|_2^2$$

This can be looked at as regression over the score funcⁿ



To compute the true scores.

$$x_0 = \frac{x_t + (1 - \alpha_t) \nabla_{x_t} \log p(x_t)}{\sqrt{\alpha_t}} \rightarrow \textcircled{1}$$

Also

$$x_0 = \frac{x_t - \sqrt{1 - \bar{\alpha}_t} \epsilon_t}{\sqrt{\bar{\alpha}_t}} \quad \epsilon_t \sim \mathcal{N}(0, I) \rightarrow \textcircled{2}$$

equating $\textcircled{1}$ & $\textcircled{2}$ & rearranging

$$\nabla_{x_t} \log p(x_t) = \frac{-1}{\sqrt{1 - \bar{\alpha}_t}} \epsilon_t$$

True score = negatively scaled noise

⇒ a DDPM trained to regress over added noise is implicitly predicting the score func

So Now we have seen 4 versions of DDPM

① Regress over True Data point

$$\|x_0(x_t) - x_0\|_2^2$$

② Regress over mean

$$\|\mu_t - \mu_0\|_2^2$$

③ Regress over noise

$$\|\hat{\epsilon}_\theta(x_t) - \epsilon_t\|_2^2$$

④ Regress over Score

$$\|\nabla_{x_t} \log p(x_t) - s_\theta(x_t)\|_2^2$$

Guided Diffusion models for Conditional Generation:

* To be able to sample from condⁿl dist.

Data : (x_0, y) where y : Conditioning Variable

eg: x_0 : Image, y : class label / text embedding

* Goal is to sample from condⁿl dist

$$p(x_0 | y)$$

question: how to modify a DDPM to sample from CondⁿL dist $p(x_0 | y)$?

Method 1: Classifier Guidance:

* Recall DDPMs are score predictions, predicting the True Score of the un-CondⁿL (a) Marginal dist.

$$\nabla_{x_t} \log p(x_t)$$

* From a CondⁿL Sampling perspective, the goal is to predict (a) regrets over the CondⁿL score

$$\nabla_{x_t} \log p(x_t | y)$$

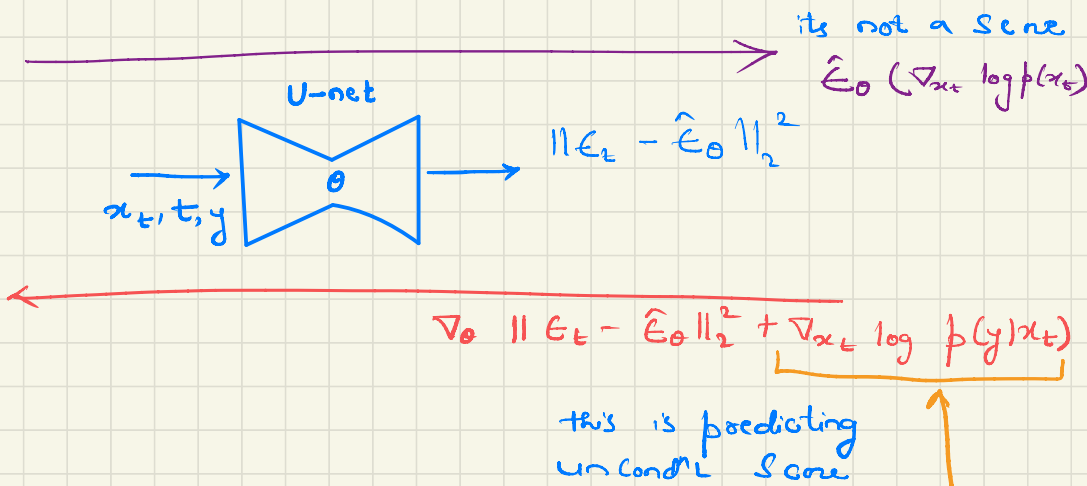
$$\nabla_{x_t} \log p(x_t | y)$$

$$= \nabla_{x_t} \log \left(\frac{p(x_t) p(y | x_t)}{p(y)} \right)$$

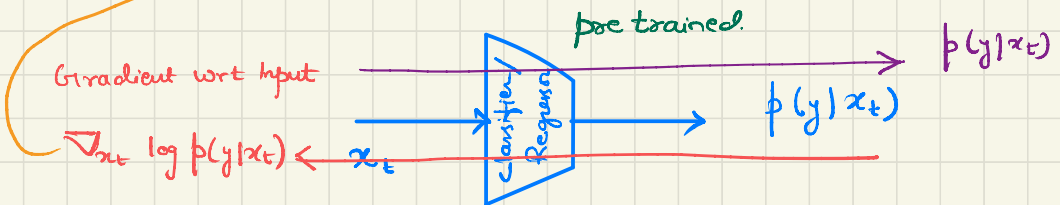
$$= \nabla_{x_t} \log p(x_t) + \nabla_{x_t} \log p(y|x_t) - \nabla_{x_t} \log p(y)$$

Now the CondⁿL score is:

$$\nabla_{x_t} \log p(x_t, y) = \underbrace{\nabla_{x_t} \log p(x_t)}_{\text{uncondⁿL score}} + \underbrace{\nabla_{x_t} \log p(y|x_t)}_{\text{classifier gradient}}$$



To get the 2nd term, train the classifier.



Classifier/Regressor is trained separately to predict y
(Class label or Embedding) from x_t

* Note that classifier is not used during sampling

so we can discard the classifier whose purpose was to guide the generation.

* The whole procedure of training the regressor on noise $\odot$ sense is dependent on how good the classifier has been trained.

* Training the classifier at diff noise levels is difficult

Method 2: Classifier free Guidance

$$\nabla \log p(x_t | y) = \nabla \log p(x_t) + \nabla \log p(y | x_t) \rightarrow \textcircled{1}$$

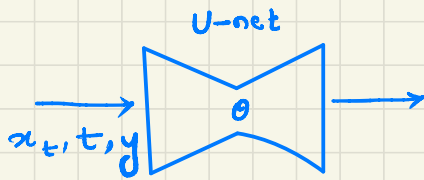
$$\nabla \log p(x_t | y) = \nabla \log p(x_t | y) - \nabla \log p(x_t) \quad \downarrow \textcircled{2}$$

$$\nabla \log p(x_t | y) \approx \nabla \log p(x_t) +$$

$$\lambda (\nabla \log p(x_t | y) - \nabla \log p(x_t))$$

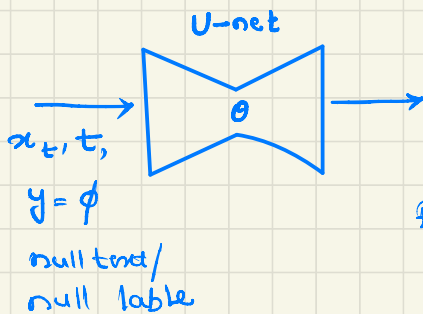
$$= \lambda \nabla \log p(x_t | y) + (1 - \lambda) \nabla \log p(x_t)$$

λ : hyper parameter



Requires over CondⁿL score $\nabla_{x_t} \log p(x_t | y)$

This is another copy of the above.



Requires over $\nabla_{x_t} \log p(x_t)$

Now before the gradient step add both with appropriate "1" as given.

Note that both are some NN

& during inference need to give "y" $\in$ every time step "t"

Latent Diffusion Models

* Basic idea: Build DMs on the latent space induced by another encoder-decoder model

* Handling DMs in high dim is difficult.

* So we use another encoder-decoder model and do the diffusion process on the latent

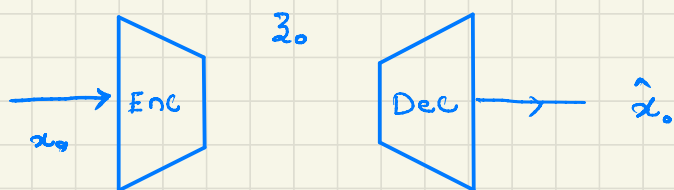
* Given data $x_0 \in \mathbb{R}^d \sim p(x_0)$

Step 1: Learn latent representation

$$z_0 \in \mathbb{R}^k \quad k \ll d$$

using Encoder-Decoder model

Eg: Construct VQ-VAE to learn a latent space



Let E_ϕ & D_θ represent pretrained encoder-decoder respectively

* E_ϕ, D_θ is trained using original data x_0

② some other dataset similar to x_0

- * All the datapoints in $\mathbb{R}^d$ are projected on to latent space as $z_0 = E_{\phi^*}(x_0)$
 - * A DDPM is constructed on the space of z_0
 - * During Inference, to generate a novel point from $p(x_0)$, first sample a novel point from the latent space with reverse diffusion z_{novel} and $x_{\text{novel}} = D_{\phi^*}(z_{\text{novel}}) \sim p(x_0)$
 - * DM is constructed on a lower dim space 'k'
 - * This is called as Stable diffusion.
-

Denoising Diffusion Implicit Models - DDIMs.

- * Limitations of DDPM:
 - Slow in sampling ($T=1000$)
 - Can't be used for deterministic posterior inference.
- * Forward process is stochastic

$$x_t = \sqrt{\alpha_t} x_0 + \sqrt{1 - \alpha_t} \epsilon_t \quad \epsilon_t \sim N(0, I)$$
- ⇒ Given an x_0 , running the forward process multiple times leads to different x_T (final latent rep for x_0)